Sub-Daily Hydration Forecasting and Lip-Based Dehydration Detection for Personalized Mobile Health Monitoring

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***Abstract*—This paper introduces a multimodal, sub-daily hydration intelligence framework for personalized mobile health monitoring. Existing hydration management applications exhibit three persistent limitations: static daily intake targets that fail to capture intra-day physiological variation; generic, population-averaged recommendations lacking genuine individualization; and exclusive reliance on self-reported data with no objective biometric assessment of real-time dehydration status. To address these limitations, the proposed system performs sub-daily water intake prediction within 4-hour windows using an XGBoost pipeline trained on a purpose-built dataset from 50 participants, achieving 87.4% accuracy (95% CI: 85.2–89.5%) and AUC-ROC of 0.921 (95% CI: 0.903–0.939). A database-driven recommendation engine integrates risk classification, ambient temperature, self-reported symptoms, medical conditions, and lip scan results to generate individualized guidance. Lip-image-based dehydration detection via a fine-tuned MobileNetV2 CNN achieves an F1-score of 0.83 (95% CI: 0.79–0.87). Explainability is provided through SHAP and Grad-CAM. All components are delivered as a Flutter mobile application. Ethics approval and informed consent were obtained from all participants prior to data collection. A usability study ($N = 48$) yielded a System Usability Scale score of 81.3 (SD $= 7.2$), classified as *Excellent*.**

***Keywords*—hydration management, dehydration detection, mobile health, lip image analysis, explainable AI**

I. Introduction

Water constitutes approximately 60% of adult body mass [21] and serves as the foundational medium for thermoregulation, nutrient transport, cognitive function, joint lubrication, and renal filtration. Despite its physiological primacy, chronic mild dehydration, defined as a body water deficit of 1–2%, is estimated to affect up to 75% of the global adult population on a daily basis, driven largely by inadequate real-time awareness and limited access to continuous monitoring tools [1], [21]. Even at this subclinical threshold, mild dehydration has been shown to reduce cognitive performance by up to 10–15% in controlled studies [1], as well as physical endurance, mood regulation, and metabolic efficiency [2].

The landscape of digital hydration tools has expanded considerably in recent years. Applications including WaterMinder, Hydro Coach, Plant Nanny, Daily Water Tracker, and Aqualert [3] demonstrate clear user demand for hydration guidance; however, a review of these widely-used tools reveals three consistent limitations. First, all existing systems prescribe a single aggregate daily water intake target, typically derived from body weight via a static formula, neglecting the substantial intra-day variation in hydration requirements driven by fluctuating activity levels, ambient temperature, and time-dependent physiological rhythms. Second, recommendations are applied at the population level rather than being calibrated to each individual user's real-time physiological state, concurrent symptoms, or environmental context. Third, all existing consumer hydration tools rely entirely on self-reported intake data, with no objective physiological measurement to verify actual dehydration status.

This paper presents a unified system addressing all three limitations through: (1) sub-daily 4-hour-window water intake prediction; (2) multi-dimensional, condition-sensitive personalized recommendations; and (3) lip-image-based dehydration detection via a fine-tuned MobileNetV2 convolutional neural network.

II. Literature Review

Hydration monitoring has been studied from clinical, technological, and machine learning perspectives. Shirreffs [7]

established that hydration status can be assessed through observable physiological markers such as urine color, skin turgor, and oral mucosa condition, and that fluid requirements vary with body weight, activity level, and ambient temperature, indicating that a single fixed daily target is insufficient for accurate individual guidance. Armstrong et al. [1] and Riebl and Davy [2] further demonstrated that even mild fluid deficits of 1–2% body weight are associated with measurable reductions in cognitive performance, physical endurance, and metabolic function, underscoring the practical importance of continuous and responsive monitoring.

On the consumer application side, Garg et al. [3] evaluated several widely used hydration tracking tools, including WaterMinder, Hydro Coach, Plant Nanny, Daily Water Tracker, and Aqualert. These applications provide scheduled reminders and manual intake logging against a static daily target derived from body weight. While reminders were found to improve adherence, none of the reviewed tools adjusted guidance in response to changes in activity, environmental conditions, or user health status, and none incorporated any objective measurement of physiological hydration state.

In the domain of predictive modelling, Ahmed et al. [5] applied a gradient-boosted ensemble to wearable sensor data and reported 89% accuracy for daily dehydration risk classification. However, the model produced binary daily-level outputs and was validated exclusively on elite athletes, limiting its generalisability to broader populations and everyday use. Chen and Guestrin [4] introduced XGBoost as a scalable and regularised tree-boosting framework well suited to structured health datasets, while Berner and La Lande [6] noted that hybrid systems combining machine learning with rule-based clinical logic can maintain interpretable and condition-specific outputs in low-data settings.

Computer vision methods have also been explored for non-invasive hydration assessment. Khosravi et al. [9] trained a convolutional neural network to estimate skin turgor from camera images as a dehydration proxy, supporting the feasibility of camera-based physiological estimation. Zhang et al. [8] applied the MobileNetV2 architecture [10] to tongue image classification for traditional medicine diagnosis, demonstrating that lightweight convolutional models can extract clinically relevant visual features at low computational cost. With respect to explainability, Lundberg and Lee [11] proposed SHAP for feature-level prediction attribution, and Selvaraju et al. [12] introduced Grad-CAM for spatially localised explanations in convolutional networks. Both have been applied in clinical imaging contexts such as diabetic retinopathy screening [13] and chest radiograph diagnosis [14], where interpretable outputs were found to support clinician trust.

Taken together, the reviewed work reveals several recurring gaps. Existing consumer hydration applications operate at a fixed daily granularity, rely entirely on self-reported data, and do not adapt to individual health conditions or real-time environmental context [3]. Predictive models for hydration risk have been validated on narrow population groups and do not provide sub-daily quantitative intake estimates [5]. Camera-based dehydration assessment has focused on skin and tongue imagery rather than lip condition, despite clinical recognition of lip appearance as a reliable hydration marker [7]. Explainability methods have not been integrated into consumer hydration tools [11], [12]. The system presented in this paper addresses these gaps through sub-daily 4-hour intake prediction, condition-sensitive personalised recommendations, lip-image-based dehydration detection, and integrated explainability via SHAP and Grad-CAM, delivered within a single mobile platform. Table I summarises how these capabilities compare with existing approaches.

TABLE I: Comparison with Existing Hydration Management Systems

Capability	**Existing Apps**	**Proposed**
Prediction granularity	Daily total	Next 4 hours
Personalization	Generic	Condition matching
Objective sensing	None	Lip image analysis
Explainable AI	None	SHAP + Grad-CAM
Weather adaptation	None	Open-Meteo
Primary dataset	Public	Purpose-built

†Assessment based on systematic review of WaterMinder, Hydro Coach, Plant Nanny, Daily Water Tracker, and Aqualert [3].

III. Methodology

The proposed system brings together three independently designed components within a unified mobile platform: a machine learning pipeline for sub-daily hydration forecasting, a computer vision module for lip-based dehydration detection, and a context-aware recommendation engine. The following subsections detail the architecture binding these components, the data collection process underpinning model training, the design of each AI model, and the mobile application delivering them to end users.

A. System Architecture

The system is structured as a three-tier client-server architecture, as illustrated in Fig. 1. A Flutter mobile application (Presentation Tier) communicates via JWT-authenticated HTTPS with a FastAPI Python REST API (Application Tier) hosting the ML inference engine. A PostgreSQL relational database forms the Data Tier. Models are lazily loaded at first inference and remain resident in memory, supporting horizontal scalability and independent component versioning. This design enables tight coupling between prediction, sensing, and recommendation components, allowing real-time interaction across modalities.

The `/tracker/dashboard` endpoint computes a personalized daily water intake goal based on established clinical guidelines [7], [15]. The base daily requirement is estimated at 35 ml per kilogram of body weight, consistent with standard adult hydration references [7], [22]. This baseline is subsequently adjusted according to four factors: (i) height, where taller individuals receive a proportionally higher allowance; (ii) biological sex, with males assigned a higher baseline than females, reflecting physiological differences in body water composition [15]; (iii) age group, where older adults receive a modestly reduced target due to lower total body water fraction [7]; and (iv) ambient temperature, where the target is

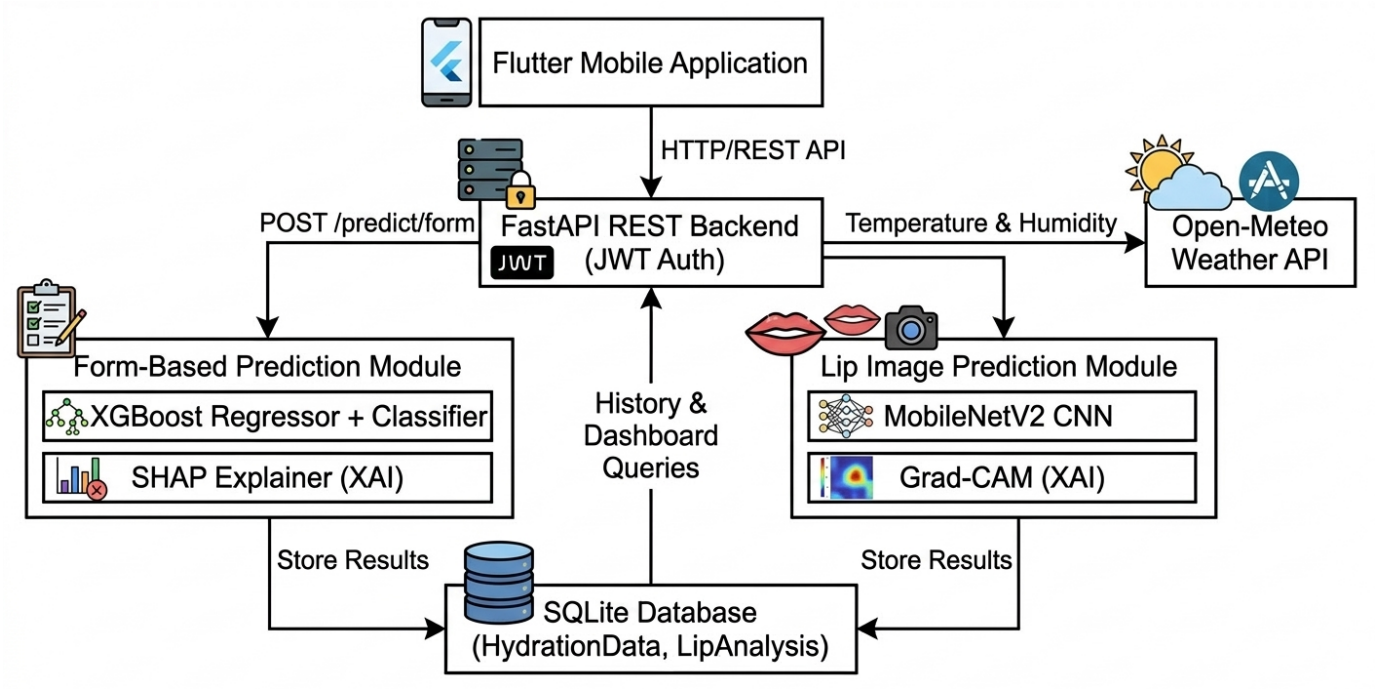

Fig. 1: System Architecture: Three-Tier Client-Server Design.

increased incrementally for conditions above 25 °C to account for elevated sweat losses [15]. These corrections are applied as fixed additive terms (in litres) derived from published reference ranges, yielding a daily intake goal G (litres), formulated as an interpretable physiological model aligned with clinical hydration guidelines:

$$G = 0.035\,w + \delta_h + \delta_g + \delta_a + \delta_t \tag{1}$$

where w is body weight (kg); $\delta_h \in \{0,\, 0.1,\, 0.2\}$ L is the height correction (0 L for height below 160 cm; 0.1 L for 160–175 cm; 0.2 L above 175 cm); $\delta_g \in \{0,\, 0.3\}$ L is the sex correction (0 L for females, 0.3 L for males); $\delta_a \leq 0$ is the age correction (up to -0.2 L for adults aged over 55); and $\delta_t \geq 0$ is the temperature correction (up to $+0.5$ L for ambient temperatures above 32 °C) [7], [15], [22]. The additive structure ensures transparency and modular adjustment of individual physiological and environmental factors, supporting explainable personalization within the system.

B. Primary Data Collection

Ethics: Conducted under SLIIT Ethics Committee approval no. SLIIT-FGS-REC-2023. Written informed consent (and parental consent for participants aged 15–20) was obtained from all participants. All data were anonymized, stored in access-controlled environments, and used exclusively for research purposes.

The hydration form dataset was constructed from fifty participants across five age strata (15–20 to 56–70 years), each completing six personalized Google Form submissions daily over two weeks. Submissions captured demographic constants, time slot, activity level, water intake, urine color on a three-level scale (Clear, Yellow, Dark) [7], and self-reported symptoms (dizziness, fatigue, headache). Ambient meteorological data (temperature, humidity, heat index) were retrospectively appended via the Open-Meteo ERA5-Land reanalysis API [18], matched to submission timestamp and participant location. After quality filtering, 4,820 usable records were retained from a theoretical maximum of 6,000.

The lip image dataset comprised 2,140 labeled images assembled from curated Google Images (Stage 1) and a dedicated smartphone collection campaign with standardized capture instructions (Stage 2). Two independent expert annotators applied labels (Cohen's $\kappa = 0.81$, almost perfect agreement), yielding 1,192 dehydrated and 948 normal samples.

C. AI Model Design

1) XGBoost Hydration Risk Classification and Regression

The pipeline employs XGBoost [4], a gradient-boosted tree ensemble that minimizes the following regularized objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i,\, y_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2 \tag{2}$$

where ϕ denotes the set of all model parameters; $\mathcal{L}(\phi)$ is the total regularized loss over all training instances; i indexes the training samples; $\hat{y}_i$ is the predicted output and y_i is the ground-truth label for instance i; $l(\hat{y}_i, y_i)$ is the task-specific loss function (cross-entropy for the classifier, mean squared error for the regressor); k indexes the trees in the ensemble; f_k is the k-th tree; $\Omega(f_k)$ is the regularization term penalizing the complexity of tree f_k; T is the number of leaves in tree f; w is the vector of leaf output weights; $\gamma \geq 0$ is the minimum loss reduction required to allow a further split; and $\lambda \geq 0$ is the L2 regularization coefficient applied to the leaf weights.

Fig. 2: Prediction output screen: 4-hour water need (0.8 L), Lip Hydration Score (20/100, Dehydrated), and SHAP-derived reasoning cards showing the top-three contributing factors.

Five derived physiological features are constructed prior to training: Body Mass Index (BMI); Heat Index (Steadman approximation [17]); Water Deficit (estimated requirement minus current intake); Sweating Factor (activity encoding $\times$ temperature deviation above 25 °C); and Hydration Efficiency Ratio (intake normalized by body weight). A `ColumnTransformer` applies `StandardScaler` to numerical features and `OneHotEncoder` to categorical features.

Two models are trained on an 80/20 stratified split with 5-fold cross-validation: a *classifier* (Normal / Mild / High Dehydration) and a *regressor* (litres recommended for the next 4 hours). SMOTE oversampling [19] is applied within training folds only, preventing data leakage into the held-out evaluation set.

SHAP TreeExplainer [11] computes per-prediction Shapley vectors; the top-three contributors are formatted as natural language reasoning cards, enabling user-facing explainable AI through interpretable reasoning of model outputs, as illustrated in Fig. 2.

2) MobileNetV2 Lip Dehydration Detection

MobileNetV2 [10] serves as the feature extraction backbone via inverted residual blocks with linear bottlenecks:

$$\mathbf{h}_{l+1} = \mathbf{P}_l \cdot \sigma(\mathbf{D}_l \cdot \sigma(\mathbf{E}_l \cdot \mathbf{h}_l)) \tag{3}$$

where $\mathbf{h}_l$ is the input feature map at layer l; $\mathbf{E}_l$, $\mathbf{D}_l$, $\mathbf{P}_l$ are the expansion, depthwise convolution, and projection matrices respectively; and σ denotes the ReLU6 activation function. Before inference, MediaPipe Face Landmarker [20] localizes the lip region and extracts four handcrafted domain features: Crack Severity Score (edge density), Redness Ratio (HSV pixels in dry/chapped hue range), Surface Roughness (LBP standard deviation), and Lip Detection Confidence. The frozen ImageNet backbone produces a 1,280-dimensional feature vector; a trainable classification head (ExpertLipModel: $1280 \to 512 \to 256 \to 128 \to 2$) trained for 30 epochs (Adam, $\mathrm{lr} = 10^{-3}$, cosine annealing) maps this to a binary softmax output. A $+0.15$ post-softmax correction is applied toward the dehydrated class when the Crack Severity Score exceeds 30, a threshold determined empirically on the validation set to account for the clinical significance of visible lip cracking as a reliable dehydration marker [7].

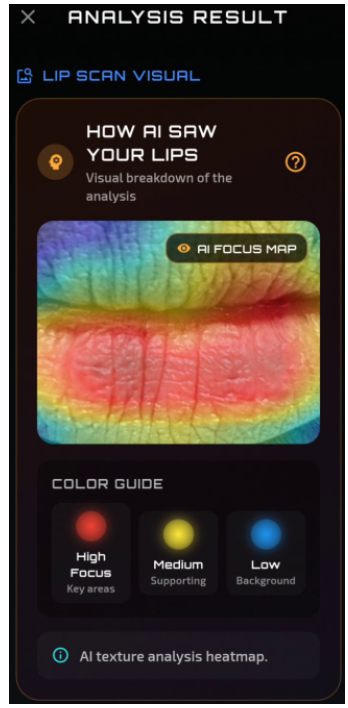

Fig. 3: Grad-CAM AI Focus Map on a representative lip image. Red = High Focus (key texture regions), yellow = Medium, blue = Low/Background, consistent with [12].

A continuous Hydration Score (HS) is computed as:

$$\mathrm{HS} = \begin{cases} 60 + (C \times 35) & \text{if Normal} \\ 45 - (C \times 35) & \text{if Dehydrated} \end{cases} \tag{4}$$

where $C \in [0, 1]$ is the softmax confidence score assigned to the predicted class by the MobileNetV2 classification head, and HS ranges from 0 to 100 with higher values indicating better hydration status.

Grad-CAM [12] heatmaps targeting the final convolutional block are overlaid in the mobile UI (Fig. 3), providing visual explainability of model attention and allowing users to interpret the physiological basis of dehydration predictions.

D. Personalized Recommendation Engine

A database-driven engine conditions recommendations on the intersection of classified risk level, ambient temperature band, actively reported symptoms, and lip scan result. Suggestions are stored as condition-predicate records in `HydrationSuggestion`, enabling clinical staff to manage guidance without code deployment. The engine distinguishes between three risk tiers (Normal, Mild Dehydration, High Dehydration) and cross-references each with temperature bands (below 25 °C, 25–32 °C, above 32 °C) and the binary lip scan result, yielding a structured recommendation matrix. Urgent alerts are visually distinguished from general behavioral guidance in the mobile UI, as shown in Fig. 4, to ensure actionable prioritization for the user. This component operationalizes the integration of predictive outputs and physiological sensing into actionable, context-aware guidance.

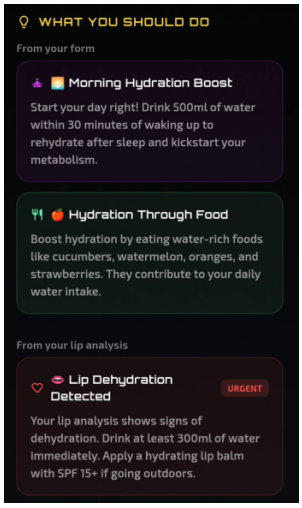

Fig. 4: Personalized recommendation screen. Guidance is conditioned on risk classification, temperature, symptoms, and lip scan result. Urgent lip-dehydration alerts (red) are distinguished from general behavioral guidance.

E. Mobile Application – Flutter Frontend

The Flutter application provides eight screens organized around two primary workflows: the *Full Assessment Flow* (form submission, stabilization timer, lip scan, combined results) and the *Quick Access Flow* (individual form or lip scan). Push notification reminders are scheduled to promote consistent hydration behavior across all six daily windows.

IV. Results and Discussion

This section presents the experimental setup, quantitative performance results for both AI components, usability evaluation findings, and a discussion of limitations.

A. Experimental Setup

The system was evaluated across two independently constructed primary datasets. The hydration form dataset comprised 4,820 records from 50 participants; an 80/20 stratified split yielded 3,856 training and 964 held-out test records, with 5-fold cross-validation for model selection. The lip image dataset comprised 2,140 labeled images partitioned into 1,712 training and 428 test samples. Performance was assessed using classification accuracy, macro-averaged F1-score, AUC-ROC, and 95% confidence intervals (1,000-iteration bootstrap). MAE and RMSE were used for regression evaluation, supplemented by R^2. A usability evaluation was conducted with 48 participants using the System Usability Scale (SUS).

B. Form-Based Prediction Results

The XGBoost classifier achieved 87.4% accuracy (95% CI: 85.2–89.5%), macro-F1 of 85.0%, and AUC-ROC of 0.921 (95% CI: 0.903–0.939) on the 964-record held-out test set. The regressor attained MAE = 0.18 L and RMSE = 0.26 L ($R^2 = 0.74$), within the ± 0.5 L tolerance considered clinically acceptable for a 4-hour window [15]. Inference latency was <50 ms for both models, confirming suitability for real-time mobile deployment. Full results are presented in Table II.

TABLE II: XGBoost Model Performance (Test Set, $N = 964$)

Metric	Classifier	Regressor
Accuracy	87.4%	—
Precision (macro)	85.2%	—
Recall (macro)	84.8%	—
F1 (macro)	85.0%	—
AUC-ROC	0.921	—
MAE	—	0.18 L
RMSE	—	0.26 L
R^2	—	0.74
Inference time	<50 ms	<50 ms

C. MobileNetV2 Lip Detection Results

The fine-tuned MobileNetV2 achieved 82.1% overall accuracy on the 428-sample test set. For the clinically critical dehydrated class, precision was 80.7%, recall 85.3%, and F1 = 0.83 (95% CI: 0.79–0.87). The balanced F1 = 0.81 for the normal class confirms no systematic bias toward either class. AUC-ROC was 0.876, reflecting strong discriminative capacity. The model ($\sim$12 MB) is served via FastAPI and completes inference within 180–220 ms end-to-end, meeting latency requirements for interactive mobile use. Results are presented in Table III.

TABLE III: MobileNetV2 Lip Detection Performance (Test Set, $N = 428$)

Metric	Value
Overall Accuracy	82.1%
Precision (Dehydrated)	80.7%
Recall (Dehydrated)	85.3%
F1 (Dehydrated)	0.83
F1 (Normal)	0.81
AUC-ROC (binary)	0.876
Model size	$\sim$12 MB
Inference time	180–220 ms

D. Usability Evaluation

A two-week field study ($N = 48$) yielded a mean SUS score of 81.3 (SD = 7.2), classified as *Excellent* on the Bangor Adjective Rating Scale [16]. Among participants, 87% reported that SHAP reasoning cards increased their trust in system recommendations; 79% found Grad-CAM lip heatmaps helpful for understanding the basis of assessments; 91% preferred the 4-hour prediction format over a standard daily goal; and push notification compliance averaged 68% across all hydration reminder windows. Qualitative feedback indicated that the visual explainability features substantially reduced perceived complexity of the AI-driven outputs, suggesting that transparent model explanations contribute to user engagement and adherence in consumer health applications.

E. Discussion

The XGBoost classifier achieved 87.4% accuracy and AUC-ROC of 0.921, which compares favorably with the 89% accuracy reported by Ahmed et al. [5] for daily-level dehydration risk classification; however, the present system additionally provides quantitative 4-hour intake estimates (MAE = 0.18 L, $R^2 = 0.74$) rather than binary risk labels, representing a meaningful increase in actionable specificity for everyday use. The regressor R^2 of 0.74 indicates that approximately three-quarters of the variance in intake requirements is explained by the model features, which is considered acceptable for

a short-horizon physiological prediction task where residual variance reflects genuine individual variability [15]. The MobileNetV2 lip detection model achieved F1 = 0.83 for the dehydrated class, which is consistent with performance reported in analogous visual health assessment tasks such as skin turgor classification [9], and demonstrates that camera-based biometric dehydration assessment is feasible in a consumer mobile context without dedicated sensors. The usability score of 81.3 (Excellent) and the strong preference for sub-daily prediction format (91%) suggest that increased prediction granularity is valued by users and may support better adherence compared to conventional daily-goal applications [3]. The integration of SHAP and Grad-CAM explainability was associated with increased user trust (87%) and comprehension (79%), consistent with findings from clinical AI studies indicating that interpretable outputs reduce perceived complexity and improve acceptance [13], [14]. Several limitations of this study should be noted. The dataset encompasses 50 participants from a single geographic region over a two-week period, and generalization to diverse climatic contexts requires larger-scale longitudinal validation. The lip image dataset partially relies on images sourced from the web, which lack verified clinical ground truth. The $+0.15$ post-softmax correction for high crack severity was determined empirically on the validation set; formal statistical optimization on an independent cohort is required to confirm its generalizability. Variability in smartphone camera quality occasionally reduces lip analysis confidence below the reliable classification threshold. The recommendation engine currently employs manually curated condition rules, and the system does not yet integrate passive sensing modalities such as wearable accelerometers or heart rate monitors, which could further enrich the physiological feature space.

V. Conclusion

This paper presented a multimodal, sub-daily hydration intelligence framework that integrates predictive modeling, visual sensing, and explainable AI within a unified mobile platform. The system demonstrated practical usability across a diverse participant group and offers meaningful value as an everyday hydration awareness tool, particularly for individuals whose fluid needs vary with activity, health conditions, or environment. The proposed approach demonstrates a shift from static hydration tracking toward real-time, explainable, and personalized hydration intelligence systems. These findings are subject to the constraints of a single-region dataset, partial reliance on web-sourced lip images, and empirically calibrated model components, all of which present clear directions for future work.

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